

Day-6 Wireless Standard & Configuration

This laboratory report outlines the practical tasks executed on a home wireless network environment using a JioFiber Gateway gateway alongside the NetSpot diagnostic application. The focus centers around analyzing frequency interference, mitigating path congestion via channel shifting, securing wireless networks, and planning operational filtering rules.

Task 1: Set up a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.

Action Taken & Methodology: The home gateway network was accessed via its localized administrative control panel of the router.

Identified Default Parameters:

- **Initial 2.4 GHz Service Set Identifier (SSID):** JioFiber-CKED2
- **Initial 5 GHz Service Set Identifier (SSID):** JioFiber-CKED2_5G
- **Initial Operational 2.4 GHz Channel:** Channel 3 (Factory Setting)
- **Initial Operational 5 GHz Channel:** Channel 157 (Factory Setting)

Network » Wireless » Access Points									
Access Points Profiles WPS MAC Filter Channel Spacing									
Access Points Record(s) Search									
Status	AP Name	Frequency	SSID	Broadcast SSID	Security	Profile Name			
✓	ap1	2.4 GHz	JioFiber-CKED2	✓	WPA2	Jio_1			
✗	ap2	2.4 GHz	Jio_2	✓	WPA2	Jio_2			
✗	ap3	2.4 GHz	Jio_3	✓	WPA2	Jio_3			
✓	ap4	5 GHz	JioFiber-CKED2_5G	✓	WPA2	Jio_4			
✗	ap5	5 GHz	Jio_5	✓	WPA2	Jio_5			
✗	ap6	5 GHz	Jio_6	✓	WPA2	Jio_6			

The SSID was changed from default to personal names, as shown in the screenshots along with fresh password.

Access Points

Profiles

WPS

MAC Filter

Channel Spacing

Access Points

Record(s) Search

Status	AP Name	Frequency	SSID	Broadcast SSID	Security	Profile Name
✓	ap1	2.4 GHz	srikant_24	✓	WPA2	Jio_1
✗	ap2	2.4 GHz	Jio_2	✓	WPA2	Jio_2
✗	ap3	2.4 GHz	Jio_3	✓	WPA2	Jio_3
✓	ap4	5 GHz	Srikant_5G	✓	WPA2	Jio_4
✗	ap5	5 GHz	Jio_5	✓	WPA2	Jio_5
✗	ap6	5 GHz	Jio_6	✓	WPA2	Jio_6

Task 2: Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks. Hint: Look for overlapping channels and pick one with the least overlap.

Action Taken & Analysis: A complete spectral analysis was launched using the NetSpot tool to identify the RF layout. The diagnostic logs exposed extreme overcrowding on mid-band frequencies due to overlapping adjacent networks running on non-standard configurations.

RF Optimization Strategy:

1. 2.4 GHz Mitigation: Spectral scanning highlighted that Channel 6 and Channel 11 were congested by dense neighboring access points (e.g., 'Crgpm', 'Mehtas601', 'Airtel_sanj_6616') registering high RSSI metrics up to -68 dBm. Channel 1 registered only highly attenuated signatures (fainter than -83 dBm). The main access point profile was switched to Channel 1 with a restricted channel width of 20 MHz to avoid wideband splatter.

☐	Airtel_gaur_4G	C4:9A:31:8F:4A:C6		-	-	-96	-84	-87		2.4	9	20	-	WPA2 Personal		50 s ago
☐	HARI OM	B4:A7:C6:51:4E:45		-	-	-96	-82	-86		2.4	9	20	-	WPA2 Personal		10 m 16 s ...
☐	Airtel_aironetouch	24:43:E2:B7:A1:58		-	-	-96	-86	-88		2.4	10 - 1	40	-	WPA2 Personal		14 s ago
☐	JioFiber-EQa5b	B4:A7:C6:C2:02:EE		-	-	-96	-74	-81		2.4	10	20	-	WPA2 Personal		17 s ago
☐	Mehtas601	3C:84:6A:26:60:22		-77	22	-96	-59	-73		2.4	10	20	-	WPA2 Personal		3 s ago
☐	Mehtas601_EXT	B4:B0:24:DA:6B:3A		-82	16	-96	-72	-83		2.4	10	20	-	WPA2 Personal		3 s ago
☐	MRUGANK 4G	B4:A7:C6:9E:36:F2		-	-	-96	-76	-85		2.4	10	20	-	WPA2 Personal		1 m 53 s a...
☐	W-48JB250	08:A6:F7:33:89:ED		-	-	-96	-77	-84		2.4	10	20	-	WPA2 Personal		3 m 21 s a...
☐	Airtel_puni_8260	0E:36:23:CF:CE:99		-	-	-96	-85	-85		2.4	11	20	-	WPA2 Personal		21 m 7 s a...
☐	Airtel_sanj_6616	B4:A7:C6:1A:9A:D9		-72	28	-96	-62	-75		2.4	11	20	-	WPA2 Personal		3 s ago
☐	Airtel_Tu Tera Dekh...	A0:91:CA:53:2B:2A		-	-	-96	-94	-94		2.4	11	20	-	WPA2 Personal		15 m 28 s ...
☐	Airtel_venk_4278	A0:91:CA:66:6D:1A		-84	14	-96	-82	-86		2.4	11	20	-	WPA2 Personal		3 s ago
☐	AP_2929430662	44:2A:8F:7C:02:AA		-	-	-96	-88	-88		2.4	11	20	-	WPA2 Personal		21 m 7 s a...
☐	BhagwanC	C2:10:07:A0:DA:D0		-	-	-96	-90	-90		2.4	11	20	-	WPA2 Personal		21 m 10 s ...
☐	DARSHANAM TWIN T...	BA:DD:71:AE:8F:14		-	-	-96	-84	-89		2.4	11	20	-	WPA2 Personal		5 m 9 s ago
☐	E90000220847335725	CE:34:5F:AD:34:DB		-	-	-96	-90	-90		2.4	11	20	-	WPA2 Personal		28 m 2 s ...
☐	OPPO A6 Pro 5G k33v	EE:D6:71:6B:28:09		-	-	-96	-80	-82		2.4	11	20	-	Unknown		29 m 6 s ...
☐	OPPO A78 5G	BE:AC:AF:B1:D0:D1		-	-	-96	-84	-84		2.4	11	20	-	WPA2 Personal		10 m 16 s ...
☐	Ram_Prasad	7A:10:07:B9:2B:40		-	-	-96	-86	-89		2.4	11	20	-	WPA2 Personal		56 s ago
☐	Samarth4G	F0:ED:B8:81:E1:2A		-	-	-96	-68	-77		2.4	11	20	-	WPA2 Personal		26 s ago
☐	Sayan	3E:F8:7E:54:80:C0		-	-	-96	-86	-89		2.4	11	20	-	WPA2 Personal		8 s ago
☐	vivo Y33T	8A:A6:1B:8CD3:A7		-	-	-96	-82	-82		2.4	11	20	-	WPA2 Personal		27 m 53 s...

2. 5 GHz Mitigation: The 5 GHz spectrum configuration showed the gateway default operating on Channel 157, which clashed directly with localized hidden casting hardware operating with an expansive 80 MHz layout. The router's frequency controller was shifted down to Channel 44 (5.22 GHz) at 80 MHz channel spacing, targeting clear, non-interfering spectral whitespace.

Task 3: Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.

Action Taken & Security Architecture: Using the system's Profile Configuration parameters, network access configurations were structurally updated to reinforce structural security and ensure clear identity separation across hardware interfaces.

Structural Modifications Applied:

- **Profile 1 (2.4 GHz Airflow) Remapping:** The primary profile 'Jio_1' was updated to use custom network name **srikant_2.4**. Security settings were explicitly bound to **WPA2-PSK (Pre-Shared Key)** utilizing advanced **Counter Mode Cipher Block Chaining Message Authentication Code Protocol (CCMP)** encryption.
- **Profile 4 (5 GHz Airflow) Remapping:** The primary high-speed profile 'Jio_4' was renamed to **Srikant_5G**, locking down access parameters via matching WPA2-PSK/CCMP protocols to ensure uniform structural integrity across both internal distribution channels.

Basic Configuration 5 GHz

Mode

ac only

Channel Spacing

80MHz

Current Channel

44 - 5.22GHz

Channel

Auto

Auto

36 - 5.18GHz

40 - 5.2GHz

44 - 5.22GHz

48 - 5.24GHz

149 - 5.745 GHz

153 - 5.765 GHz

157 - 5.785 GHz

161 - 5.805 GHz

Maximum Transmit Power

Cancel

Save

Wireless Profiles Configuration

Profile Name

Jio_4

SSID

Srikant_5G

[Length: 1 - 32]

Broadcast SSID

ON

Security

WPA2

Encryption

CCMP

Authentication

PSK

WPA Password

[Length: 8 - 63]

Tasks 4 Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result. Hint: You can find your device's MAC address in the network settings.

Access Points Profiles WPS **MAC Filter** channel Spacing

MAC Filter

AP Name AP1/AP4

ACL Policy Status Allow

Cancel Save

MAC Address

Record(s) Search + Add new

Show 10 entries

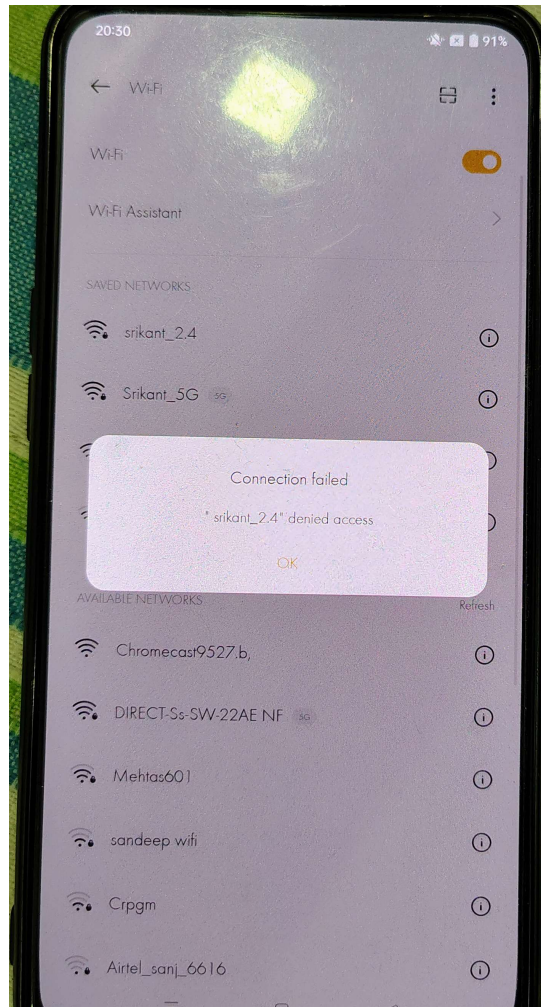
MAC Address

4c:6f:9c:f7:95:7d

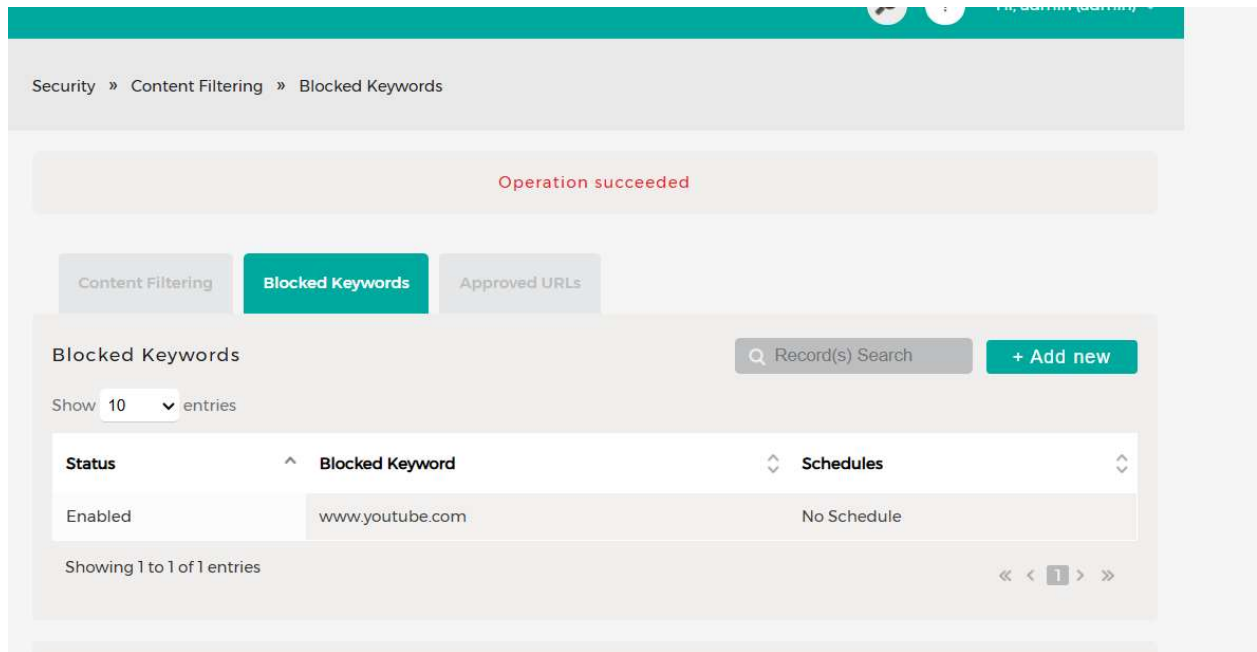
Showing 1 to 1 of 1 entries

To verify MAC Filtering, the Android device's hardware MAC address (4c:6f:9c:f7:95:7d) was added to the Access Control List under a **Allow** policy on the JioFiber Gateway. Upon attempting to reconnect another device to the srikant_2.4 SSID, the connection was immediately rejected by the router.

As captured in the verification screenshot, the mobile device displays a explicit terminal message: **"Connection failed: 'srikant_2.4' denied access."** This confirms successful layer-2 filtering, where the router blocks network association requests based purely on hardware identifiers, regardless of the password supplied.



Task 5: Create an Access Control List (ACL) on your router to block access to a specific website (for example, www.youtube.com) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.



Goal - Block websites (: //youtube.com) connection from local Wi-Fi network devices through our router by creating an Access Control List (ACL) rule in our firewall and test from our client device.Equipment Used/Jobs Performed:- Logged into the JioFiber router's dashboard by accessing 192.168.29.1 from our browser.- Open Security > Content Filtering > Blocked Keywords Screen.- Enabled Content Filtering Status option.- Entered a new row with ://youtube.com at the end of domain in our firewall table listed under any "Enabled" "No Schedule" (permanent) option and hit Save.

RESULT/Findings: I tried accessing <https://youtube.com> from my laptop connected to the Wi-Fi. Internet connection was working fine except for that specific link. The router firewall rule did the job as it dropped connection requests to that website. Browser did not show YouTube page but showed a network connecting failure page with a message that read -"Connect to the internet. You're offline. Check your connection." This message is from YouTube's server indicating that the domain was unable to resolve because it got blocked by our router.



Connect to the internet

You're offline. Check your connection.

Retry

